

Noise & Edges and Contours (Part 1)

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- Revise: Histogram Equalization
- Revise: Increase/Decrease Gamma
- Noise
- Edges and Contours

Revise: Histogram Equalization

- Let $P_I(\gamma + 1)$ is the CDF of I type int8
- Call J is the remap of I following this equation, in which b means the b color channel of I

$$J(r, c, b) = 255 \times P_I(I(r, c, b) + 1) \quad (1)$$

- Exercise: Let I (3 bits) $\in R^{4 \times 4}$:

$$I = \begin{pmatrix} 0 & 0 & 0 & 4 \\ 1 & 1 & 1 & 5 \\ 1 & 2 & 2 & 7 \\ 2 & 2 & 2 & 7 \end{pmatrix}$$

- Calculate image J using equation (1).
- Calculate histogram of J

Revise: Increase/Decrease Gamma

- Increase Gamma

$$J(r, c) = 255 \times \left[\frac{I(r, c)}{255} \right]^{\frac{1}{\gamma}} \text{ in which } \gamma > 1.0$$

- Decrease Gamma

$$J(r, c) = 255 \times \left[\frac{I(r, c)}{255} \right]^{\frac{1}{\gamma}} \text{ in which } \gamma < 1.0$$

- While taking picture (during capture), noise may occur
- Noise? Errors, degradations in pixel values
- Examples of causes:
 - Focus blurring
 - Blurring due to camera motion
- Additive model for noise:

$$H * I + \text{Noise}$$

- Removing noise called **Image Restoration**
- Image restoration can be done in: *Spatial domain*, or *Frequency domain*

Types of Noise

- Type of noise determines best types of filters for removing it!!
- **Salt and pepper noise**: Randomly scattered black + white pixels
- Also called **impulse noise**, **shot noise** or **binary noise**
- Caused by sudden sharp disturbance



(a) Original image



(b) With added salt & pepper noise

*Courtesy
Allasdair McAndrews*

Types of Noise

- **Gaussian Noise**: idealized form of white noise added to image, normally distributed:

$$I + \text{Noise}$$

- **Speckle Noise**: pixel values multiplied by random noise

$$I(1 + \text{Noise})$$



(a) Gaussian noise



(b) Speckle noise

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Allasdair McAndrews*

Types of Noise

- **Periodic Noise:** caused by disturbances of a periodic nature
- Salt and pepper, gaussian and speckle noise can be cleaned using spatial filters
- Periodic noise can be cleaned using frequency domain filtering (later)

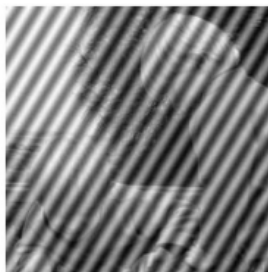
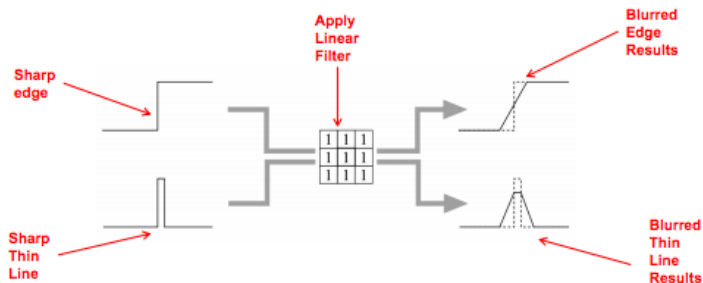


Figure 5.3: The twins image corrupted by periodic noise

*Courtesy
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Non-Linear Filters

- Linear filters blurs all image structures points, edges and lines, reduction of image quality (**bad!**)
- Linear filters thus not used a lot for removing noise



Using Linear Filter to Remove Noise?

- Example: Using linear filter to clean salt and pepper noise just causes smearing (not clean removal)
- Try non-linear filters?



(a) 3×3 averaging



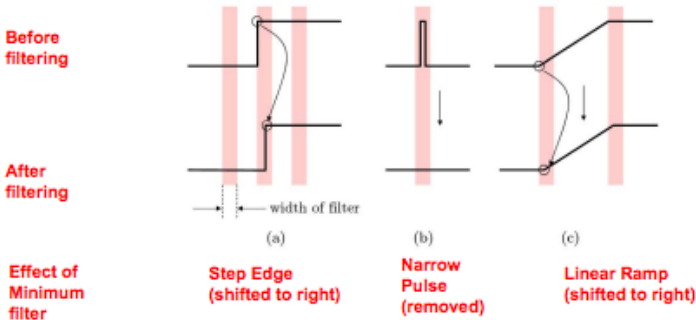
(b) 7×7 averaging

Non-Linear Filters

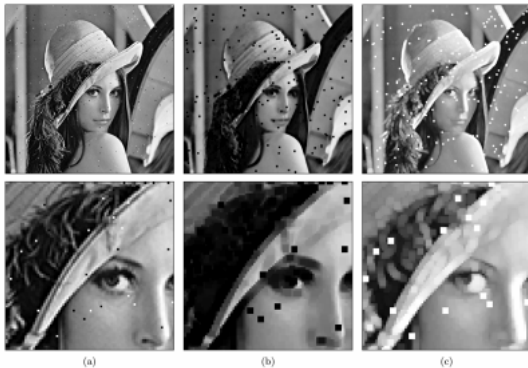
- Pixels in filter range combined by some non-linear function
- Simplest examples of nonlinear filters: Min and Max filters

$$I'(u, v) \leftarrow \min\{I(u + i, v + j) | (i, j) \in R\}$$

$$I'(u, v) \leftarrow \max\{I(u + i, v + j) | (i, j) \in R\}$$



Non-Linear Filters



- (a): Original Image with Salt-and-pepper noise
- (b): Minimum filter removes bright spots (maxima) and widens dark image structures
- (c): Maximum filter (opposite effect): Removes dark spots (minima) and widens bright image structures

Median Filter

- Much better at removing noise and keeping the structures

$$I'(u, v) \leftarrow \text{median}\{I(u + i, v + j) | (i, j) \in R\}$$

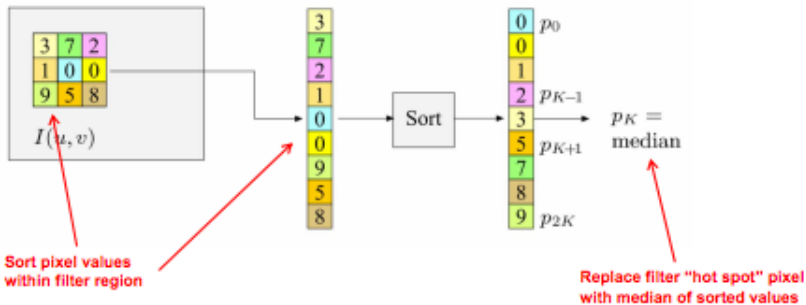
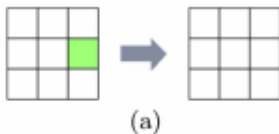
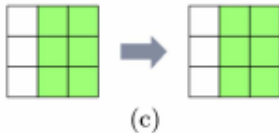
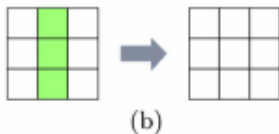


Illustration: Effects of Median Filter

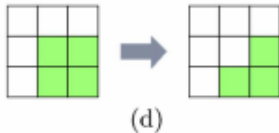
Isolated pixels
are eliminated



Thin lines
are eliminated



A step edge is
unchanged



A corner is
rounded off

Non-Linear Filters



- (a): Original Image with Salt-and-pepper noise
- (b): Linear filter removes some of the noise, but not completely. Smears noise
- (c): Median filter salt-and-pepper noise and keeps image structures largely intact. But also creates small spots of flat intensity, that affect sharpness

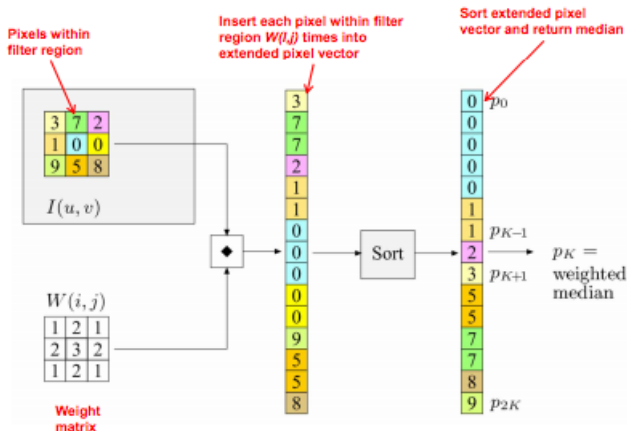
Weighted Median Filter

- Color assigned by median filter determined by colors of “the majority” of pixels within the filter region
- Considered robust since single high or low value cannot influence result (unlike linear average)
- Median filter assigns weights (number of “votes”) to filter positions

$$W(i,j) = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

- To compute result, each pixel value within filter region is inserted $W(i,j)$ times to create extended pixel vector
- Extended pixel vector then sorted and median returned

Weighted Median Filter



- Note: assigning weight to center pixel larger than sum of all other pixel weights inhibits any filter effect (center pixel always carries majority)!!

Weighted Median Filter

- More formally, extended pixel vector defined as

$$Q = (p_0, \dots, p_{L-1}) \text{ of length } L = \sum_{(i,j) \in R} W(i,j)$$

- For example, following weight matrix yields extended pixel vector of length 15 (sum of weights)

$$W(i,j) = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

- Weighting can be applied to non-rectangular filters
- Example: cross-shaped median filter may have weights

$$W^+(i,j) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

An Outlier Method of Filtering

- Algorithm by Pratt, Ref: Alasdair McAndrew
- Median filter does sorting per pixel (computationally expensive)
- Alternate method for removing salt-and-pepper noise
 - Define noisy pixels as outliers (different from neighboring pixels by an amount $> D$)
- Algorithm:
 - Choose threshold value D
 - For given pixel, compare its value p to mean m of 8 neighboring pixels
 - If $|p - m| > D$, classify pixel as noise, otherwise not
 - If pixel is noise, replace its value with m ; Otherwise leave its value unchanged
- Method not automatic. Generate multiple images with different values of D , choose the best looking one

Outlier Method Example

- Effects of choosing different values of D



(a) $D = 0.2$



(b) $D = 0.4$

*Courtesy
Allasdair McAndrews*

- D value too small: removes noise from dark regions
- D value too large: removes noise from light regions
- D value of 0.3 performs best
- Overall outlier method not as good as median filter

Other Non-Linear Filters

- Any filter operation that is not linear (summation), is considered linear
- Min, max and median are simple examples
- More examples later:
 - Morphological filters
 - Corner detection filters
- Also, filtering shall be discussed in frequency domain

- Digital Image Processing (CS/ECE 545), Computer Science Dept. Worcester Polytechnic Institute (WPI)